

TECHNICAL DECISION REPORT — AUTONOMOUS BUSINESS RESEARCH

Should we launch an electric scooter subscription service in Bengaluru?

GOAL INTERVIEW · HYPOTHESES · MULTI-SOURCE EVIDENCE · CHALLENGE & VERIFY · TRIAL · VERDICT

DOCUMENT ID ROPS-2026-6A6F0B	SCOPE as stated in the question	RUN DATE 2026-10-01 07:45
--	---	-------------------------------------

GENERATED FROM CITED EVIDENCE — EVERY STATEMENT TRACES TO A VERBATIM SOURCE QUOTE

VERDICT: LEANING FAVOURABLE (UNCERTAIN)
51% FAVOURABLE · RANGE 25%-78%

SOURCES 31 20 useful	CLAIMS 604 591 independent	EVIDENCE 32 labels weighed	CHALLENGED 8 re-researched	HEARINGS 1 evidence court	TOKENS 81543 LLM use
-----------------------------------	---	---	---	--	-----------------------------------

DOCUMENT CONTROL

ITEM	VALUE
Prepared by	ResearchOps autonomous agent (openai/gpt-oss-120b)
Run ID	6a6f0be752
Citation verification	13 supported · 13 partial · 16 struck
Red-team survival	FAILS

Probabilities are computed by a transparent log-odds belief engine from independent evidence clusters; the language model reads and classifies evidence but never sets a number.

01

EXECUTIVE TECHNICAL SUMMARY

§01 verdict

01.1 PRINCIPAL FINDINGS

- 01Operating costs could stay below the target INR 1,500 per scooter per month thanks to lower fuel expenses and existing charging-swap infrastructure, but users say battery charging costs may be 3.8 times higher than expected. [c324, c603, c596]
- 02Regulatory approval appears achievable within weeks under standard two-wheeler registration, yet challengers note recent statements suggesting additional permits could delay approval beyond three months. [c123, c120, c126, c129, c89, c535]
- 03Competitor micromobility services currently hold a combined market share below 15%, leaving room for a subscription model, although some evidence reports dominant players with 35% share and scaling challenges. [c542, c526, c369, c536]

01.2 RECOMMENDED ACTIONS

- 01Partner with charging-swap providers to keep operating costs at or under INR 1,500 per scooter per month. [c603, c324, c219]
- 02Complete standard two-wheel registration within 30 days of vehicle delivery before launch. [c123, c126, c129]

WHAT WOULD CHANGE THE VERDICT

KIND	FACT / ASSUMPTION	VERDICT WITHOUT IT	FLIPS?
assumption	Our team can set up charging stations and perform routine maintenance at a cost of ≤INR 500 per scooter per month.	41%	YES
assumption	Daily commuters aged 18-35 own or have easy access to a smartphone and are comfortable with digital payments.	43%	YES
assumption	Local traffic authorities will treat subscription-based scooters the same as privately owned two-wheelers for registration and parking.	44%	YES

02

HYPOTHESES AND ODDS

§02 belief engine

ID	HYPOTHESIS	ODDS	RANGE	FOR / AGAINST
h1	At least 8% of daily commuters aged 18-35 in the target zones are willing to pay a monthly subscription fee of INR 2,500 for an electric scooter.	38%	14%–70%	1 / 1
h2	The average monthly operating cost per scooter (maintenance, charging, insurance, parking) can be kept below INR 1,500.	55%	30%–77%	5 / 2
h3	Regulatory approval for a subscription-based electric scooter service can be obtained within 3 months without requiring additional permits beyond standard two-wheeler registration.	59%	29%–84%	7 / 3
h4	Competitor micromobility services (e.g., bike-share, e-scooter rentals) have a combined market share below 15% in the selected zones, leaving room for a subscription model.	47%	20%–77%	2 / 4
h5	Customers prefer a fixed monthly fee over per-trip charges for electric scooter usage.	68%	44%–85%	7 / 0
h6	The average churn rate for the first six months will be under 20% once the service is launched.	45%	17%–77%	0 / 0

STATED ASSUMPTIONS

ID	ASSUMPTION	LIKELIHOOD
a1	Daily commuters aged 18-35 own or have easy access to a smartphone and are comfortable with digital payments.	85%
a2	Our team can set up charging stations and perform routine maintenance at a cost of ≤INR 500 per scooter per month.	85%
a3	Local traffic authorities will treat subscription-based scooters the same as privately owned two-wheelers for registration and parking.	75%
a4	Existing micromobility operators are not planning aggressive expansion or price wars in the pilot zones within the next year.	60%

03

CHALLENGE AND VERIFICATION

§03 claim lifecycle

STATUS	CLAIM	CHALLENGE	VERIFIER
INSUFFICIENT_EVIDENCE	Monthly electric scooter rentals are gaining popularity due to: c235 · HYPOTHESIS	The claim that monthly electric scooter rentals are gaining popularity lacks quantitative data or trend analysis to substantiate the growth assertion. MEDIUM	nothing in 3 follow-up page(s) bears on the claim
INSUFFICIENT_EVIDENCE	In contrast, renting a bike monthly can be significantly cheaper, with all-inclusive pricing covering maintenance, insurance, and taxes. c202 · INFERENCE	Stating that monthly bike rentals are "significantly cheaper" than ownership is vague without a cost comparison model that includes all variables. HIGH	nothing in 3 follow-up page(s) bears on the claim
INSUFFICIENT_EVIDENCE	In this comprehensive guide, we will explore everything you need to know about monthly bike rental in Bangalore, including its benefits, top rental services, pricing, documents required, rules, and important tips. c199 · INFERENCE	The guide claims to be comprehensive but provides no evidence of covering all relevant regulations or regional variations in Bangalore. LOW	nothing in 3 follow-up page(s) bears on the claim
INSUFFICIENT_EVIDENCE	Full insurance, helmet, and free maintenance included. c219 · FACT	The promise of full insurance, helmet, and free maintenance may be limited by fine print or exclusions that are not disclosed. MEDIUM	nothing in 3 follow-up page(s) bears on the claim
PARTIALLY_SUPPORTED	Offers Bounce Infinity electric scooters on long-term rent with subscription models. c215 · HYPOTHESIS	Mentioning Bounce Infinity scooters on long-term rent does not address availability across models or geographic coverage, which could limit the claim's applicability. LOW	related evidence found but not decisive (support 1%, contradiction 45%)
INSUFFICIENT_EVIDENCE	Monthly plans starting from ₹3,000/month (depending on the model). c218 · INFERENCE	The starting price of ₹3,000/month is presented without context of model differences, deposit requirements, or hidden charges. MEDIUM	no follow-up sentence was about the same subject
INSUFFICIENT_EVIDENCE	Owning a bike involves a substantial upfront cost and recurring expenses such as insurance, maintenance, fuel, and EMI payments. c201 · INFERENCE	The statement that owning a bike involves substantial upfront and recurring costs oversimplifies financing options and may ignore low-cost models or subsidies. MEDIUM	nothing in 3 follow-up page(s) bears on the claim

INSUFFICIENT_EVIDENCE	“The unit economics and maintenance benefits of switching to electric are very encouraging, and we already have a trial with around 100 electric scooters,” Gupta said. c324 · HYPOTHESIS	The quote about unit economics and a trial of 100 electric scooters is anecdotal and lacks independent verification of cost savings or scalability. HIGH	nothing in 3 follow-up page(s) bears on the claim
-----------------------	--	---	---

NUMERIC CONTRADICTIONS BETWEEN SOURCES

MEASURE	SUBJECT	VALUES	CLAIMS
fleet/users	Drivezy	[30.0, 15.0]	c83, c374

04

EVIDENCE COURT

§04 per-claim hearings

QUALIFIED LAST YEAR, BENGALURU-BASED VOGO RAISED USD 100 MILLION FROM OLA TO ADD 100,000 SCOOTERS TO ITS CURRENT FLEET OF 4,000.

Exhibit c384 alone confirms the claim, but no independent evidence corroborates the \$100 M investment or the 100,000-scooter expansion, so the claim is only partially supported.

ROLE	ARGUMENT	CITES	CHECK
JUDGE	The court will hear the evidence regarding Vogo's alleged funding and fleet expansion. • Opening statement: Proceed with the claim and exhibits.		no citation
CLERK	The claim and exhibits are presented. • Claim text: Vogo raised USD 100 million from Ola to add 100,000 scooters to its 4,000-scooter fleet. • Exhibit list: Exhibits c66, c82, c374, c52 are also on file.	c384, c66, c82, c374, c52	supported
PROSECUTION	The claim lacks corroborating evidence and is contradicted by market context. • No funding proof: No independent source confirms a \$100 M investment by Ola. • Demand drop: Industry faced a severe demand drop, making such expansion unlikely. • Fleet size mismatch: Other Indian scooter-sharing fleets are far smaller, casting doubt on a 100 k increase.	c66, c82, c52	supported
DEFENSE	The claim itself stands as the sole documented evidence of the transaction. • Claim source: Exhibit c384 records the \$100 M raise and fleet plan. • No direct refutation: Other exhibits discuss different companies or markets, not Vogo's funding.	c384, c66, c82, c374, c52	supported
JUDGE	Ruling based on the admissible exhibits. • Limited support: Only the claim itself (c384) supports the allegation; no external verification is provided. • Contradictory context: Other exhibits highlight market weakness but do not directly disprove the specific transaction.	c384, c66, c82, c52	unsupported

05

RED-TEAM AUTOPSY

§05 second-order audit

SURVIVAL: FAILS
SEVERITY SCORE 34.5 (7 MEDIUM, 3 CRITICAL, 3 HIGH); THRESHOLDS 4 / 10 / 18; ANY CRITICAL FINDING CAPS THE VERDICT AT WEAKENED.

SEVERITY	AUDITOR	FINDING	ACTION
CRITICAL	Assumption Auditor	Unverified assumption can flip the verdict: Daily commuters aged 18-35 own or have easy access to a smartphone and are comfortable wit Reversing it moves the verdict to 43%.	Ask the decision-maker or find evidence for this assumption.
CRITICAL	Assumption Auditor	Unverified assumption can flip the verdict: Local traffic authorities will treat subscription-based scooters the same as privately own Reversing it moves the verdict to 44%.	Ask the decision-maker or find evidence for this assumption.

CRITICAL	Logic Auditor	Assuming lower operating costs despite battery cost spike The verdict claims operating costs can stay below INR 1,500 per scooter per month, yet evidence shows battery charging expenses rose 3.8 times the expected level ([c596]). The analysis jumps from a single cost outlier to a blanket conclusion that costs will remain low.	Model full cost structure including worst-case battery charging scenarios and test sensitivity of pricing.
HIGH	Evidence Auditor	16 struck and 13 partial of 42 cited statements The entailment verifier found many report or trial statements not fully supported by their quotes.	Rewrite or drop the struck statements.
HIGH	Logic Auditor	Conflicting market share evidence leads to unjustified revenue projection The decision assumes ample room for a subscription model based on a claimed combined competitor market share below 15% ([c542]), while another source reports a dominant player holding 35% ([c369]). This contradictory data is used to infer market size without reconciling the discrepancy, resulting in an unsupported revenue outlook.	Conduct an independent market share analysis to establish the true competitive landscape before finalising financial models.
HIGH	Methodology Auditor	Absence of primary demand validation and comparison groups The evidence set relies on competitor announcements, regulatory notes, and anecdotal cost figures, but lacks primary user research or a control group comparing subscription versus outright ownership preferences.	Run a pilot study or survey to measure willingness to pay and subscription uptake versus existing rental models.
MEDIUM	Temporal Auditor	21 cited claim(s) are stale for their kind Claims the conclusion relies on are old relative to how fast their kind of fact changes (prices, competitors).	Re-check these figures against current sources.
MEDIUM	Contradiction Auditor	1 numeric conflict(s) between sources are unresolved fleet/users of Drivezy: [30.0, 15.0]	Find which figure is current and from whom.
MEDIUM	Data Quality Auditor	7 statistic(s) are cited without a sample size or measurer Percentages and figures appear without who measured them or on how many.	Find the underlying study or mark the figure as unverified.
MEDIUM	Completeness Auditor	Hypothesis h6 rests on 0 independent evidence cluster(s) The average churn rate for the first six months will be under 20% once the service is launched.	Research this hypothesis directly.
MEDIUM	Alternative Hypothesis Auditor	Low adoption rates of electric two-wheelers could suppress demand Evidence indicates overall adoption of electric two-wheelers remains low ([c166]), which could explain limited market share for existing players and may be the primary barrier, rather than pricing or subscription model suitability.	Investigate consumer adoption drivers and barriers through primary surveys to gauge realistic demand.
MEDIUM	Completeness Auditor	Regulatory approval timeline uncertainty not fully quantified While some sources suggest approval within weeks ([c120], [c126]), others note potential additional permits that could extend the timeline beyond three months ([c89]). The decision does not account for this variance nor its impact on launch scheduling.	Map the full regulatory pathway, identify all required permits, and build a contingency timeline into the project plan.
MEDIUM	Completeness Auditor	Infrastructure scalability and competition for charging slots omitted Evidence mentions existing charging-swap infrastructure ([c603]) and a trial of 100 electric scooters ([c324]), but does not assess whether this infrastructure can support a larger subscription fleet or how competition for slots may affect service reliability.	Evaluate current charging network capacity and plan for scaling or partnership agreements before fleet expansion.

06 COMPARISON MATRIX \$06 cited cells only						
OPTION	MONTHLY PRICE	UNLOCK FEE	PER KM COST	FLEET SIZE	FUNDING	FINANCIAL LOSS
Yulu	₹2,499/month [c213]	Rs 10 per trip [c184, c309]	₹0.25/km [c236]	15,000 vehicles [c7]	\$8 M (Bajaj) + \$7 M total [c66, c479]	Rs 95 cr FY23 loss [c592]
Bounce	—	—	Rs 5/km [c306]	Target 50,000 e-scooters by year-end [c366]	\$72 M Series C + \$75 M round + \$25 M plant + \$50-75 M battery [c366, c532, c508]	—
Vogo	—	—	Rs 3/km (cheaper than Bounce) [c306]	4,000 current, aim 100,000 [c384]	\$100 M from Ola [c384]	—

07 ACTION PLAN \$07 phased roadmap

FIRST STEP
RUN A LOW-COST PILOT TO VALIDATE CHARGING-SWAP COST AND REGISTRATION COMPLIANCE.

PHASE	ACTIONS	GO / NO-GO GATE	RISKS RETIRED
Validate Core Assumptions 1 month	- Partner with a local charging-swap provider to set up 2 pilot swap stations covering a 5 km radius. - Deploy 20 scooters to a test cohort of 18-35-year-old commuters with smartphones. - Track monthly charging-swap cost per scooter and ensure it stays ≤INR 500. - Complete standard two-wheel registration for each scooter within 30 days of delivery. [c498, c215, c126, c603]	Go if average swap cost ≤INR 500 and 100% of scooters are registered within 30 days.	Charging-swap cost may exceed target (must stay ≤INR 1500 total). Regulatory compliance risk for registration timing. User digital-payment adoption risk.
Pilot Subscription Model 3 months	- Launch a 3-month subscription to the 20-scooter pilot cohort at INR 2,500 per month. - Monitor churn, average daily rides, and payment success rate. - Negotiate a formal agreement with the charging-swap partner to lock in ≤INR 1,500 per scooter per month total cost. - Engage local traffic authority to confirm subscription scooters receive same registration/parking treatment as private two-wheelers. [c215, c120, c89, c526]	Go if churn < 15%, payment success ≥95%, and authority confirms equal treatment.	Subscription pricing viability. Authority treatment risk. Total operating cost ceiling.
Scale Preparation 2 months	- Secure contracts with 3 additional charging-swap partners to cover 5 km zones city-wide. - Finalize SOPs for routine maintenance targeting ≤INR 500 per scooter per month. - Build a digital onboarding flow confirming smartphone ownership and digital-payment capability. - Create a compliance checklist confirming registration within 30 days for all fleet units. [c324, c596, c120, c603]	Go if cost model shows ≤INR 1,500 total monthly cost per scooter and compliance checklist passes 100%.	Scaling cost risk. Maintenance cost risk. Compliance scalability.

Full Launch 6 months	• Deploy 1,000 scooters across Bengaluru with partnered swap stations. • Roll out subscription plans with tiered pricing based on pilot insights. • Monitor key KPIs: cost per scooter, registration compliance, churn, and market share. • Report to investors and consider SoftBank follow-on funding if > 5% market share achieved. [c369, c526, c535, c542]	Go if average cost ≤ INR 1,500, registration compliance 100%, churn < 15%, and market share ≥ 5% after 6 months.	Full-scale economic viability. Regulatory risk at scale. Market adoption risk.
--------------------------------	--	--	--

08

UNIT ECONOMICS

\$08 per customer per month

MARGIN 2442 INR	LTV 34886 INR	LTV/CAC 69.771 ratio	PAYBACK 0.205 months	P (LTV/CAC>3) 100% 2000 trials	EVIDENCE SHARE 20% cited params
------------------------------	----------------------------	-----------------------------------	-----------------------------------	---	--

PARAMETER	LIKELY	RANGE	BASIS	NOTE
price	2500	2400–3000	EVIDENCE [c213, c218]	Monthly subscription plans advertised from ₹2,499 to ₹3,000.
variable_cost	58	40–80	ASSUMPTION	Assumes average usage 200 km/month at ₹0.25/km (₹50) plus battery swapping/charging ops ~₹8 per month; range reflects usage variance.
cac	500	300–800	ASSUMPTION	Typical customer acquisition cost for scooter-sharing in Indian metros, based on marketing spend and incentives; conservative range.
churn	0.07	0.04–0.12	ASSUMPTION	Monthly churn for subscription mobility services in emerging markets; range captures higher early-stage attrition.
fixed_cost	500000	300000–800000	ASSUMPTION	Includes staff, rent, software, and overhead for a Bengaluru operation; range reflects lean vs. scaled staffing.

SCENARIOS — WORST CASE TO BEST CASE

SCENARIO	MARGIN	LTV	LTV/CAC	PAYBACK	HOW
Worst case	2320.0	19333.333	24.167	0.345	every parameter at its unfavourable bound
Severe downside (p5)	2394.242	23834.41	40.173	0.142	5th percentile of simulated futures
Downside (p10)	2416.166	25513.305	44.082	0.155	10th percentile
Likely (p50)	2552.197	34381.722	65.998	0.205	median of simulated futures
Upside (p90)	2765.515	46990.398	97.739	0.265	90th percentile
Best case	2960.0	74000.0	246.667	0.101	every parameter at its favourable bound

SENSITIVITY (LTV/CAC SWING PER PARAMETER)

PARAMETER	AT LOW	AT HIGH	SWING
churn	122.1	40.7	81.4
cac	116.286	43.607	72.679
price	66.914	84.057	17.143
variable_cost	70.286	69.143	1.143
fixed_cost	69.771	69.771	0.0

09

SOURCE REGISTER

§09 cited sources

#	SOURCE	TYPE	CLAIMS
01	Yulu to focus on long-term rentals, add 1 lakh electric bikes by 2021 https://www.livemint.com/companies/news/yulu-to-focus-on-long-term-rentals-add-1-lakh-electric-bikes-by-2021-11596979896600.html	News / business publication	c83
02	After Tejasvi Surya asks Karnataka govt to reconsider carpooling ‘ban’, transport minister Reddy says apps need permission Bangalore News - The Indian Express https://indianexpress.com/article/cities/bangalore/transport-minister-ramalinga-reddy-on-carpooling-ban-karnataka-8965496/	News / business publication	c89
03	Karnataka Vehicle Registration: Process, Fees & Documents https://parkplus.io/blog/rto/motor-vehicles-act-and-rules-karnataka-vehicle-registration	Blog / article	c120, c123, c126, c129
04	Bike Rental Bangalore Monthly https://24fbr.gnanasamruddhi.in/bike-rental-bangalore-monthly/	Blog / article	c219
05	Do Indians Just Hate Each Other Too Much To Share Bikes? HuffPost Tech https://www.huffpost.com/archive/in/entry/bike-sharing-startups-yulu-vogo-bounce-parking-problems_in_5cdd98bde4b0b4728ba4bc98	Blog / article	c324
06	Will mobility startups keep fueling EV growth despite India’s fluctuating stance? https://kr-asia.com/will-mobility-startups-keep-fueling-ev-growth-despite-indias-fluctuating-stance	Blog / article	c369, c374
07	The electric scooter capital of the world World Economic Forum https://www.weforum.org/stories/geographies-in-depth/bengaluru-scooter-sharing-india/	Blog / article	c526, c535, c536, c542
08	Yulu’s FY23 loss widens on battery charging-related costs YourStory https://yourstory.com/2023/11/yulus-fy23-loss-widens-evs-delivery-zepto	News / business publication	c596, c603